

BBM 448 - Assignment 2

Distributed Key-Value Store with gRPC

Learning Objectives

- Finish the tutorial at <https://grpc.io/docs/languages/go/quickstart/> to understand **client-server communication** using gRPC in Go.
 - Implement **distributed data storage** with **replication and consistency** ideas.
 - Practice **protobuf definition, streaming, and concurrent request handling**.
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Problem Statement

You are to implement a **distributed key-value store** with **3 server nodes** communicating via **gRPC**.

Each node stores a local map of `key → value` pairs.

Clients can connect to **any node** to perform:

- `Put(key, value) →` Store or update a value
- `Get(key) →` Retrieve a value
- `List() →` Stream all key-value pairs

Each write (`Put`) operation must be **replicated** to the other nodes via gRPC before being acknowledged to the client.

Requirements

1. **Define the gRPC service:**
 - Methods:
 - `Put(KeyValue)` returns `(Ack)`
 - `Get(Key)` returns `(Value)`
 - `List(Empty)` returns `(stream KeyValue)`
 - Inter-node replication: servers call each other's `Replicate(KeyValue)` RPC internally.
2. **Server Implementation:**
 - Each server runs on a separate port (e.g. 5001, 5002, 5003).
 - Maintain a local `map[string]string` as storage.
 - On `Put`, write locally and send `Replicate` RPCs to peers.
 - Handle concurrent access safely (use a mutex).
3. **Client Implementation:**
 - Client connects to any node.

- Supports CLI commands:
 - put key value
 - get key
 - list
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File Structure

```
grpc-kvstore/  
├── proto/  
│   └── kvstore.proto  
├── server/  
│   └── main.go  
└── client/  
    └── main.go
```

How to Test/Run

```
# generate Go files  
protoc --go_out=. --go_opt=paths=source_relative \  
    --go-grpc_out=. --go-grpc_opt=paths=source_relative proto/kvstore.proto  
  
# start 3 servers  
go run server/main.go 5001 5002,5003  
go run server/main.go 5002 5001,5003  
go run server/main.go 5003 5001,5002  
  
# start client on any port, for example port 5001  
go run client/main.go 5001
```

Example CLI session:

```
> put name Adnan  
Stored and replicated  
> get name  
name = Adnan  
> list  
name = Adnan
```

Example CLI session 2:

```
% go run client/main.go 5001
> hi
Commands: put/get/list/exit
> get
Usage: get <key>
> get 1
1 =
> put 1 adnan
Stored and replicated
> get 1
1 = adnan
> get adnan
adnan =
> list
1 = adnan
> exit
```

Submission Guidelines

- Submit the entire `grpc-kvstore` folder as a zip file to submit.cs.hacettepe.edu.tr
- Deadline: 23.59 @ Tuesday, 28.10.2025 (no late submission, no extension)